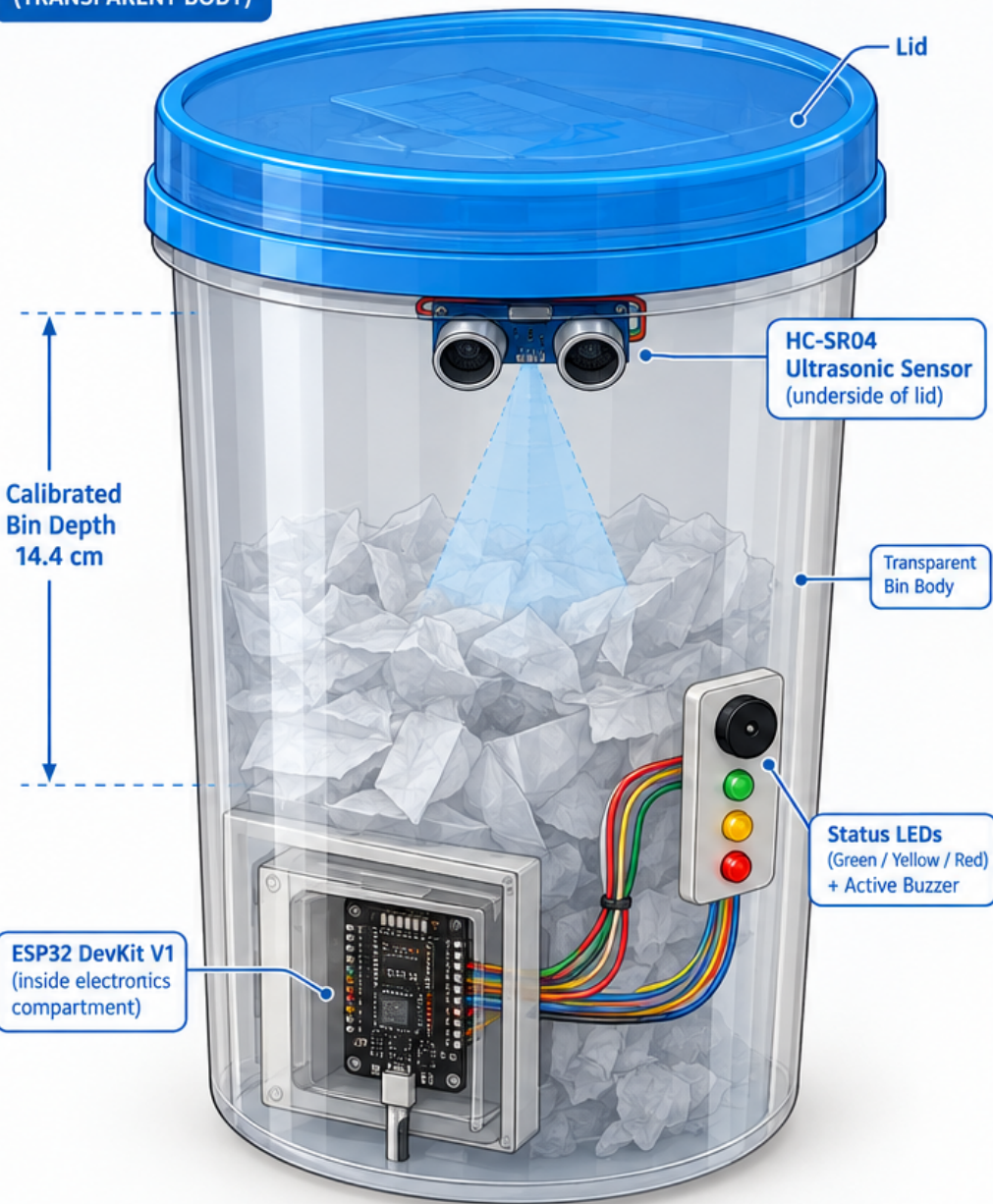


ISOMETRIC VIEW (TRANSPARENT BODY)



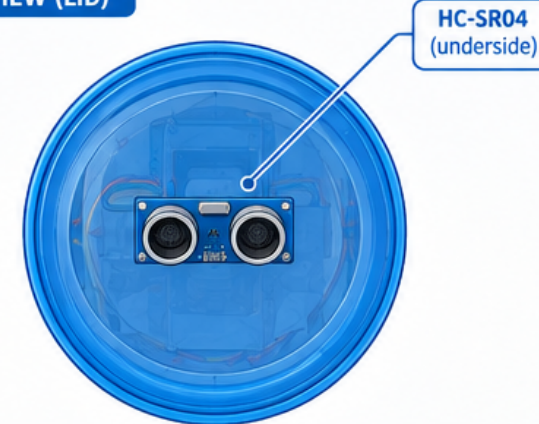
FRONT VIEW



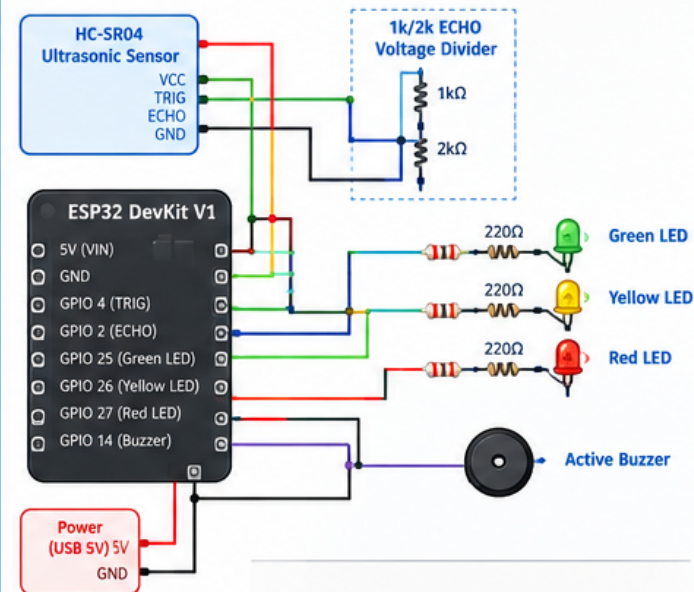
SIDE VIEW



TOP VIEW (LID)



ELECTRONICS WIRING DIAGRAM



HC-SR04

Measures distance to waste surface



ESP32 DevKit V1

Processes data, hosts Wi-Fi AP, serves dashboard



3 × Status LEDs

Green (OK)
Yellow (Half)
Red (Full)



Active Buzzer

Alerts on full bin condition



14.4 cm

Calibrated bin depth



Wi-Fi Access Point
(no internet required)



Live Dashboard
(Phone / Laptop)



Smart Infrastructure
· Urban Sanitation



SMARTBIN-1 — CAD PROTOTYPE DIAGRAM

Automated IoT Waste-Level Monitoring & Alert System

Prototype V1.0

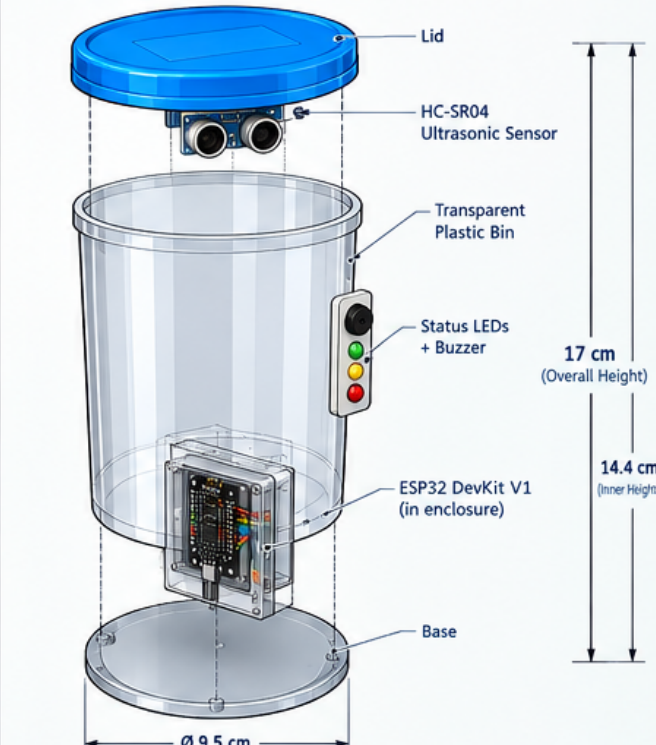
1. 3D ISOMETRIC VIEW (ASSEMBLED)



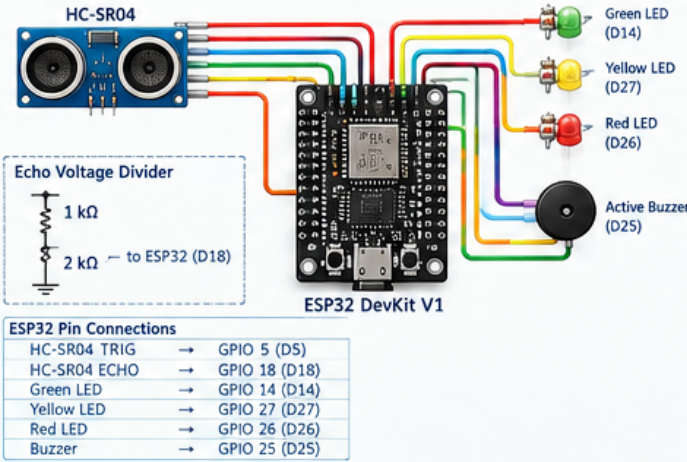
2. CAD VIEWS & DIMENSIONS



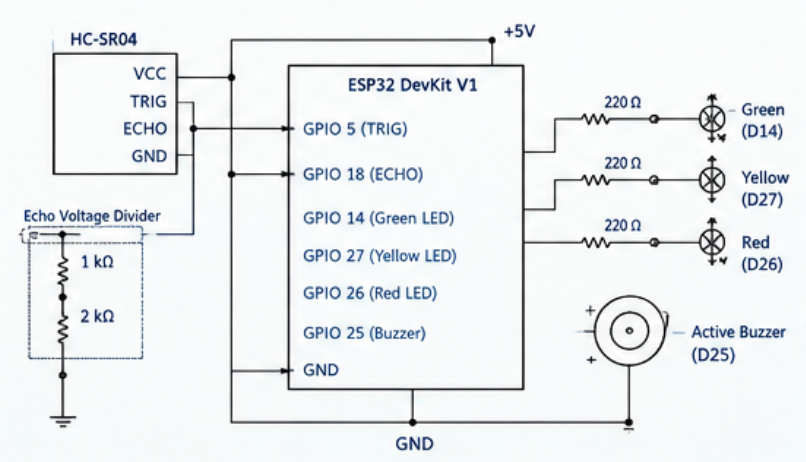
3. EXPLODED VIEW



4. WIRING DIAGRAM



5. ELECTRICAL SCHEMATIC



6. BILL OF MATERIALS

No.	Component	Model / Specification	Qty.
1	ESP32 DevKit V1	ESP-WROOM-32	1
2	Ultrasonic Sensor	HC-SR04	1
3	LED (Green)	5 mm, 2.0 V, 20 mA	1
4	LED (Yellow)	5 mm, 2.0 V, 20 mA	1
5	LED (Red)	5 mm, 2.0 V, 20 mA	1
6	Active Buzzer	5 V, 2 kHz	1
7	Resistor (LEDs)	220 Ω, 1/4 W	3
8	Resistor (Voltage Divider)	1 kΩ + 2 kΩ	1 set
9	Jumper Wires	Male-Female / Male-Male	—
10	Plastic Bin	~9.5 cm (Ø) x 17 cm (H)	1
11	Enclosure (Electronics)	Plastic box	1
12	USB Power Supply	5 V, 2 A (mobile charger)	1

7. KEY FEATURES



8. PROJECT INFO

Prototype	: V1.0
Group	: 3
Course	: MEE2014
Dept.	: CSE (B.Tech)
Year	: 2026

SmartBin-1

Automated IoT Waste-Level Monitoring & Alert System

Smart Infrastructure • Clean Environment • Better Tomorrow

Prepared by:
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Aryan Singh (25BCE10798)
VIT Bhopal University, SCOPE

Drawing No.: SB1-001
Scale: NTS
Units: mm
Date: 2026-05-30

Prototype CAD Diagram
(Concept Design)